Youyang Shang

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EDUCATION

The Ohio State University — Columbus, OH Aug 2026 – Dec 2029 (Expected)
Bachelor of Science in Mathematics, Honors theoretical track
Minor in Computer Science

TECHNICAL SKILLS

Programming: C++, C#, Java, JavaScript, Python; Unity Engine; object-oriented programming; data structures; algorithms
Mathematics: Mathematical Analysis, Statistics
Quantum: Qiskit; quantum cryptography fundamentals
Languages: Mandarin Chinese (native), English (fluent)

RESEARCH EXPERIENCE

Math Cycle Program Sep 2026 – Present
Topic: Fractal Geometry · Mentor: Ori Salim Friesen

- Study geometric patterns with self-similarity and recursive generation, including fractal dimensions and complex forms found in nature.
- Use computational simulations and mathematical proofs to analyze fractal models and investigate their properties and behavior.

LEADERSHIP

Buckeyes ACG Piano Club — The Ohio State University Starting Feb 2027 (Expected)
Founder & President

- Founding a student organization that brings together piano enthusiasts and ACG (anime, comics, games) fans through music.
- Planning piano performances, music theory tutoring, and collaborative workshops; managing club operations and recruitment.

PROJECTS & EXTRA CURRICULARS

CampusPilot — HackOHIO 2026 Oct 2026 – Present
Team Lead · Algorithm Designer

- Built a campus task-planning application with three teammates using React, TypeScript, and Python.
- Tested scheduling algorithms and verified application functionality through automated test cases.
- Developing post-hackathon quantum optimization and room-booking extensions with IBM Qiskit.

Dimensional Adventure — Autumn GameJam 2026 Sep 2026
Programmer · Prefabs, Assets & UI

- Contributed to gameplay and systems programming for a Unity 2D pixel-art GameJam project.
- Built prefabs and assets from team-provided artwork, and managed their integration.
- Designed and implemented the main menu and in-game UI elements to support intuitive navigation.

Quantum Cryptograph — Qiskit Fall Fest 2026 Nov 2026
Programmer

- Used quantum algorithms to recover encryption keys and decrypt 10 locked files in a problem-solving competition.
- Built post-processing and decryption workflows, optimized quantum circuits, and verified solutions through official tests.

AstralNote — OSU JPAS Club Nov 2026 – Present
Intern Keyboarder

- Perform keyboard accompaniment for anime and ACG music band rehearsals and live performances.
- Assist with musical arrangements across styles and collaborate with band members to ensure cohesive performances.

Updated November 2026